

Pillar B — Model Explanation and Design Rationale

1. Step-by-Step Explanation of the Proposed Model

We propose a CNN-based architecture with Convolutional Block Attention Module (CBAM) that can help classify the samples from the SOMLAP dataset. It starts with the 108 features of the input data set. These features are organized into a synthetic feature grid of 12×9 features and then resized to a three-channel representation of 96×96 image. Note that this transformation is not intended to mean that the data has its spatial structure in nature, but it is only a computational representation for CNN. Then the input is passed through four convolutional stages to get the transformed output. The different layers learn simpler pattern representations at the shallow ones and more complicated pattern representations as part of the deeper layers. The CBAM attention is added to the first three convolutional blocks to enable the network to selectively highlight useful representations. CBAM has two complementary ways of working. Channel attention narrows down which feature maps learnt are more relevant for the classification task, on the other hand, spatial attention highlights potentially important locations within the learnt feature representation. Then the extracted features are summarized by Global Average Pooling and fed into fully connected layers. Last, there is a sigmoid output layer that gives the probability that a sample is in the malware class. The architecture also includes Batch Normalization and Dropout to ensure stable training and enhance generalization. A held-out test set created with leakage-aware grouping procedures is used to evaluate the final model.

2. Rationale Behind the Design Choices

The proposed architecture is developed with a purpose or motivation, but not additional parts just to complicate the model. The feature grid is 12×9 to avoid discarding information as the data set has 108 features, and is completely possible to fit into a 12×9 array. This is a kind of representation that allows performing convolutional operations to investigate local patterns of the encoded features. But as it is a synthetic arrangement, any spatial patterns are interpreted with some caution. A CNN was chosen as the main feature extractor since it has the ability of efficient learning of hierarchical representations. As the network progresses, progressively more complex combinations of features are captured so that the network can learn more and more complex combinations. To evaluate the effectiveness of selective attention as a solution to a standard CNN, CBAM was introduced. The underlying assumption is that some feature maps learned by the system are more important than others in the classification decision. Channel attention allows the network to focus on more informative feature maps, and spatial attention offers another way of focusing attention on relevant regions of the learned representation. With the consideration of fair evaluation of the contribution of attention, the plain CNN baseline

model was designed by the same backbone and classification structure without CBAM. This will facilitate the comparison since the differences in performance are less likely to be due to changes in unrelated architectures. To facilitate flatter optimization Batch Normalization (BN) was applied and Dropout as a regularisation was applied to avoid overfitting. Global Average Pooling (GAP) was chosen over a large Flatten-based classifier in order to maintain a maximum amount of high-level information while reducing the number of parameters. Our augmentation strategy was also aimed at being conservative. Simple intensity-based perturbations were used without any transformations (flip or rotation) because the change in the position of the features may lead to an artificial change of its semantic content in the synthetic grid.

3. Honest Interpretation of the Results

While the results of the experiments are encouraging, there are lessons to be learned as well. In the first pair of matched, the CNN+CBAM model resulted in 0.961578 accuracy and 0.959021 in macro-F1 while the plain CNN model attained 0.950841 accuracy and 0.948204 in macro-F1. This represents a gain of $\sim 1.08\%$, showing that attention can help improve balanced classification performance on Macro-F1. But the improvement was not found in all of the evaluation measures. Malware precision increased but the malware recall fell from 0.941552 to 0.924085. From a real-world perspective, this resulted in fewer false positive detections, but a higher percentage of malware samples that were missed. This trade-off is especially crucial in a classification problem related to security and should not be neglected. Accordingly, we cannot consider the early improvement to be evidence that CBAM is always superior. The next ablation experiments explore further how various architectural aspects influence performance. Results from this study are also subjected to validation-based model selection, a held-out test evaluation, a paired statistical test, and grouped cross validation to explore if the conclusions drawn are generalizable across experimental splits. The explainability analysis results gotten from Grad-CAM and LIME helps to give further understanding of the model's activities on individual predictions both right and wrong predictions. However, these explanations are to be taken as a guide to model sensitivity, not causal proof. Highlighted regions represent an approximate location within the representation of the input, and do not necessarily correspond to a physically meaningful region in the original data.

Overall Conclusion

In general, the proposed CNN+CBAM approach proves that adding attention mechanisms could be useful for improving balanced classification performance. The outcomes also illustrate a crucial balance between precision and recall, not least in the context of malware identification. Hence, we conclude cautiously that this task is well suited for CBAM but its performance will vary with the evaluation criterion, realization method and specific application's constraints. Instead of going on claiming the proposed model is better in all respects, in our work we are interested in the reasoning behind the architecture of this model; we test its components using controlled experiments; and we interpret the improvements and the limitations of this model transparently.